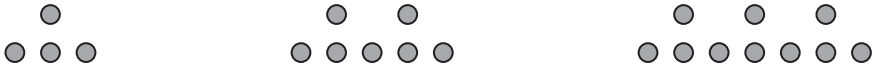
18 Here is a sequence of patterns made with counters.



pattern number 1 pattern number 2 pattern number 3

(a) Find an expression, in terms of n, for the number of counters in pattern number n.

Counters: 4, 7, 10, ... Increase by 3 each time.

1 mark
Use 3n for the constant increase

3n gives 3, 6, 9, ... Add 1: 3n + 1.

3n + 1

(2)

1 mark
Correct nth term

Bayo has 90 counters.

(b) Can Bayo make a pattern in this sequence using all 90 of his counters?
You must show how you get your answer.

$3n + 1 = 90$

$-1 \quad -1$

$3n = 89$

$\div 3 \quad \div 3$

1 mark
Set the expression equal to 90

$n = 89 \div 3 = 29.666...$, not a whole number.

No. Pattern 29 has 88 counters; pattern 30 has 91.

(2)

1 mark
A pattern number must be a positive whole number

(Total for Question 18 is 4 marks)